

Define Problem Statements

Ideation Phase

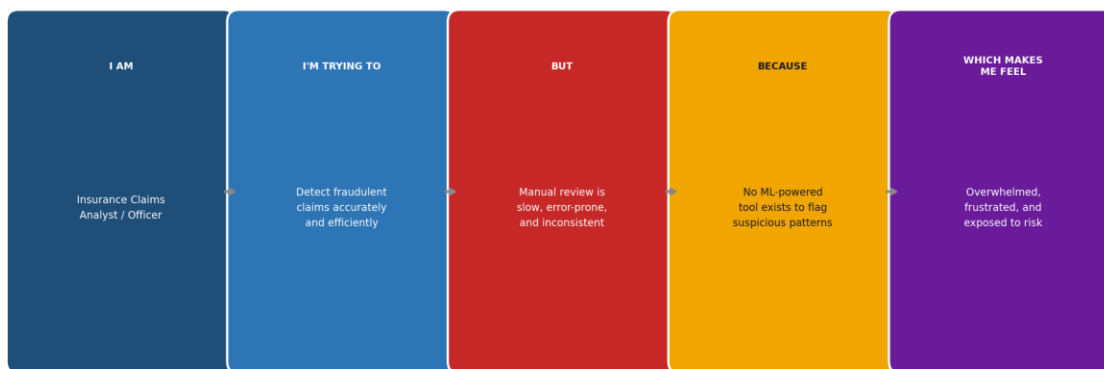
Date	07 March 2026
Team ID	
Project Name	Insurance Fraud Detection Using Machine Learning
Maximum Marks	2 Marks

Customer Problem Statement Template

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you will also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

The Customer Problem Statement framework uses the structure: I am... I'm trying to... but... because... which makes me feel — to articulate the root cause of the user's pain.

Customer Problem Statement Framework



Problem Statements — Insurance Fraud Detection

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An Automobile Insurance Claims Analyst processing 50+ claims daily	Quickly identify fraudulent vehicle damage claims and approve only legitimate ones	Manual review of claim photos, repair estimates, and customer history takes 3–4 hours per case	There is no automated system that can flag suspicious patterns across damage reports, claim amounts, and customer claim history	Overwhelmed, inefficient, and constantly worried I'm missing hidden fraud
PS-2	A Health Insurance Reviewer handling medical reimbursement claims	Detect overfilled, duplicate, or medically unnecessary treatment claims before payment	Treatment codes, billing patterns, and claim frequency are too complex and voluminous to review manually with accuracy	No intelligent pattern-recognition system exists to cross-check billing codes against treatment norms and frequency benchmarks	Frustrated, error-prone, and exposed to significant financial risk for the company

PS No.	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-3	A Property Insurance Claims Manager assessing damage claims post-disaster	Accurately validate claims related to fire, theft, or natural disaster damage against actual property value and timing	Inflated or entirely false property claims are submitted with fabricated documentation that looks authentic	No ML model exists to cross-verify reported claim values against property records, damage patterns, and historical claim behavior	Uncertain, under-resourced, and anxious about making wrong payout decisions that harm company finances
PS-4	An Insurance Company Risk Manager overseeing all claim portfolios	Get a consolidated, data-driven view of fraud risk across automobile, health, and property segments in real time	Fraud data is soiled across different teams with no unified fraud scoring dashboard or trend analytics	Legacy systems don't integrate ML predictions or provide visual fraud trend analytics across segments	Blind to emerging fraud patterns and unable to make proactive, data-driven risk management decisions

How Might We Statements

Based on the problem statements above, the following 'How Might We' challenge questions guide our solution design:

#	How Might We Statement	Linked Problem
HM W-1	How might we build an ML system that automatically scores automobile insurance claims for fraud risk in real time, saving analysts 3+ hours per day?	PS-1
HM W-2	How might we detect overfilling and duplicate health insurance claims by recognizing abnormal billing code patterns and treatment frequencies?	PS-2
HM W-3	How might we use ML to cross-reference property claim values against historical records and damage benchmarks to flag inflated claims automatically?	PS-3
HM W-4	How might we provide a unified fraud analytics dashboard that gives risk managers real-time visibility into fraud trends across all insurance segments?	PS-4